

NORWAY HOUSE, MB, Canada

WMO: 714100

Lat: 53.960N

Lon: 97.840W

Elev: 224

StdP: 98.67

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.3	(c) -33.8	(d) -40.6	(e) 0.1	(f) -36.1	(g) -38.1	(h) 0.1	(i) -33.6	(j) 8.4	(k) -15.0	(l) 7.6	(m) -15.5	(n) 1.0	(o) 300	(p) 0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.4	(c) 27.6	(d) 18.7	(e) 26.0	(f) 17.9	(g) 24.5	(h) 17.3	(i) 20.4	(j) 25.1	(k) 19.4	(l) 24.0	(m) 18.4	(n) 22.8	(o) 4.3	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.9	22.7	17.6	13.0	21.6	16.7	12.2	20.6	59.8	25.2	56.3	24.0	53.1	22.8	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.3	(b) 7.3	(c) 6.4	(e) -39.8	(f) 30.5	(g) 3.0	(h) 1.6	(i) -42.0	(j) 31.6	(k) -43.7	(l) 32.5	(m) -45.4	(n) 33.4	(o) -47.6	(p) 34.6		
(a) 8.3	(b) 7.3	(c) 6.4	(e) -39.9	(f) 22.2	(g) 3.0	(h) 1.6	(i) -42.0	(j) 23.4	(k) -43.8	(l) 24.3	(m) -45.5	(n) 25.2	(o) -47.6	(p) 26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-20.2	-17.9	-10.1	-0.1	7.9	14.8	18.1	16.7	10.9	2.9	-7.7	-16.9		
	DBStd	14.82	8.03	7.61	8.09	6.68	4.97	3.99	3.09	3.49	4.53	4.57	6.93	8.19		
	HDD10.0	4391	937	782	622	306	102	7	0	1	43	227	531	833		
	HDD18.3	6782	1196	1016	881	553	324	118	42	75	226	480	781	1092		
	CDD10.0	723	0	0	0	2	36	151	252	207	70	5	0	0		
	CDD18.3	73	0	0	0	0	1	11	36	23	3	0	0	0		
	CDH23.3	599	0	0	0	0	14	120	281	164	19	0	0	0		
	CDH26.7	92	0	0	0	0	1	19	46	24	2	0	0	0		
(14) Wind	WSAvg	3.0	2.7	2.8	3.1	3.3	3.5	3.1	2.9	3.0	3.1	3.2	2.9	2.6		
(15) Precipitation	PrecAvg	530	23	24	28	23	44	65	69	82	57	45	37	32		
	PrecMax	653	54	60	55	58	113	135	126	179	87	101	68	45		
	PrecMin	374	9	3	5	7	2	24	21	35	16	6	18	15		
	PrecStd	78	12	14	14	16	36	30	30	42	22	26	13	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	2.6	9.0	19.3	25.3	29.2	30.1	29.1	25.8	18.9	7.9	1.2		
		MCWB	0.3	1.2	4.0	10.4	14.6	19.2	20.9	19.8	17.9	12.4	5.4	-0.1		
	2%	DB	-2.7	0.1	6.2	15.7	22.0	26.2	27.8	27.0	22.6	14.5	5.5	-1.4		
		MCWB	-3.4	-0.8	2.8	8.6	12.5	17.3	18.8	18.9	16.4	10.1	3.6	-2.1		
	5%	DB	-5.4	-2.6	3.8	12.5	19.5	24.4	26.2	25.2	20.4	12.0	3.2	-3.1		
		MCWB	-5.9	-3.6	1.3	6.4	11.4	16.3	18.1	18.2	15.1	8.8	1.8	-3.6		
	10%	DB	-7.8	-6.0	1.4	9.7	17.0	22.6	24.8	23.4	18.4	9.8	1.1	-5.1		
		MCWB	-8.2	-6.9	-0.3	4.5	10.2	15.3	17.7	17.3	14.1	7.4	0.2	-5.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	1.2	4.9	11.2	15.6	21.0	22.2	21.6	19.3	13.7	5.8	0.0		
		MCDB	0.9	2.1	8.2	18.3	22.5	26.8	27.4	26.7	24.0	17.5	7.4	0.9		
	2%	WB	-3.4	-0.8	3.0	8.7	13.6	18.9	20.5	20.2	17.4	10.8	3.7	-2.0		
		MCDB	-2.8	0.0	6.0	14.8	20.0	23.8	25.1	24.7	21.2	13.5	5.2	-1.4		
	5%	WB	-5.9	-3.6	1.5	6.9	12.2	17.5	19.5	19.2	15.9	9.1	1.9	-3.7		
		MCDB	-5.4	-2.6	3.7	12.0	17.9	22.4	24.2	23.6	19.7	11.7	3.1	-3.1		
	10%	WB	-8.3	-6.8	-0.2	5.0	10.8	16.3	18.5	18.2	14.5	7.6	0.4	-5.6		
		MCDB	-7.9	-6.0	1.4	9.2	16.1	21.0	23.1	22.1	17.8	9.7	1.1	-5.1		
(35) Mean Daily Temperature Range	MDBR		9.2	11.5	12.5	11.9	11.4	11.1	10.4	9.9	9.3	7.0	7.3	8.5		
	5% DB	MCDBR	9.0	9.9	11.7	15.1	15.3	13.9	12.3	11.8	12.2	10.9	8.2	8.1		
		MCWBR	8.6	8.9	8.6	9.1	7.7	6.6	5.4	5.5	6.8	7.0	6.5	7.6		
	5% WB	MCDBR	9.0	9.7	10.5	14.2	13.2	11.5	10.5	10.2	10.9	9.8	7.6	7.8		
		MCWBR	8.6	8.7	7.8	8.8	7.3	6.1	5.5	5.6	6.7	6.6	6.2	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.253	0.278	0.334	0.344	0.371	0.390	0.360	0.324	0.293	0.264	0.233		
	taud		2.264	2.300	2.330	2.372	2.441	2.375	2.359	2.468	2.525	2.568	2.360	2.253		
	Ebn at Noon		738	863	913	896	904	877	851	861	855	815	710	675		
	Edh at Noon		57	78	95	106	105	114	114	96	80	61	52	47		
(44) All-Sky Solar Radiation	RadAvg		0.85	1.91	3.33	4.90	5.54	5.72	5.57	4.54	3.15	1.69	0.84	0.61		
	RadStd		0.09	0.13	0.27	0.39	0.48	0.45	0.34	0.36	0.33	0.20	0.08	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page